

Quantum Cryptography: Enabling Secure Data Transmission

Welcome to the world of spies and secret messages. This article focuses on the application of quantum mechanics to transmit secret messages. Wannabe cryptographers will find it an interesting read.



Traditional cryptographic methodologies rely on mathematics and have theoretical limits on computational power. There are different mathematical tricks and techniques that facilitate the factorisation of extremely large numbers into their corresponding primes, hence making cryptanalysis—the study of finding security vulnerabilities in cryptographic systems—a popular and achievable task. The RSA has done reasonably well over the last two decades and remains most popular yet resilient to hackers. Named after its three inventors, Rivest, Shamir and Adleman, according to Wikipedia, it is one of the first practicable public-key cryptosystems and is widely used for secure data transmission. Factoring a modulus is referred to as a *brute-force* attack in the case of RSA. The most efficient factoring algorithm is *general number field sieve (GNFS)*, that runs in $O(L^c)$, for ' n bit' integer; where $c < 2$. Though GNFS is far from polynomial bounds and in current state doesn't possess much threat to RSA. Several planned attacks to exploit its mathematical aspects are possible—like low exponent (both private and public), blinding, chosen cipher text, cycle attacks,

etc. The computational power of electronic devices is increasing day by day, with even Moore's law approaching obsolescence. Moreover, the onset of research and development on quantum computing makes current cryptographic methods prone to cryptanalysis.

Quantum cryptography takes encryption and decryption to a different level. It brings physics into action rather than relying too much on mathematical computations and assumptions. The US government backed DARPA's (Defense Advanced Research Projects Agency) quantum network is the world's first fully functional quantum cryptographic network running between DARPA, Harvard University and Boston University. Let us now take a ride on quantum mechanics.

Perception and simulation

Qubit: The basis of quantum cryptography is a qubit (a quantum bit). The term 'quantum' is derived from the word *quanta*, which means a packet of energy. This packet contains photons, and these photons constitute a qubit. Thus, quantum computing and quantum